

Shoes Sales

May 13, 2026

1 Import libraries

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Read a CSV file called shoes sales

```
[3]: df = pd.read_csv(r"C:\Users\Nkhosikhona Mpungose\Downloads\shoes_sales_dataset.
↪CSV")
df
```

```
[3]:
```

	Sale_ID	Date	Brand	Shoe_Type	Color	Country	Sales_Channel	\
0	S1	2025-11-24	Nike	Boots	Blue	UK	Online	
1	S2	2025-03-13	Skechers	Boots	Grey	USA	Mall	
2	S3	2025-08-05	Nike	Running	White	UK	Mall	
3	S4	2025-11-05	New Balance	Casual	Green	UAE	Mall	
4	S5	2025-10-07	Adidas	Formal	Grey	France	Online	
..	...	...	...	...	...	...	...	
995	S996	2025-06-06	Puma	Sneakers	Red	India	Online	
996	S997	2025-08-14	Adidas	Casual	Green	France	Online	
997	S998	2025-02-13	Skechers	Running	White	India	Online	
998	S999	2025-05-28	Puma	Boots	Black	France	Retail Store	
999	S1000	2025-08-30	Puma	Casual	Green	India	Online	

	Price_USD	Units_Sold	Revenue_USD
0	112.40	4	449.60
1	239.16	4	956.64
2	191.04	2	382.08
3	161.70	1	161.70
4	64.32	14	900.48
..	...	...	...
995	50.15	6	300.90
996	231.81	2	463.62
997	60.10	4	240.40
998	239.05	1	239.05

```
999      128.12      1      128.12
```

```
[1000 rows x 10 columns]
```

2 Checking for any duplicates

2.0.1 The code below shows that the data has no duplicates

```
[52]: df.duplicated().sum()
```

```
[52]: np.int64(0)
```

```
[ ]:
```

3 Check number of unique values

```
[92]: df.nunique()
```

```
[92]: Sale_ID      1000
      Date        342
      Brand        6
      Shoe_Type    6
      Color        6
      Country      7
      Sales_Channel 3
      Price_USD    979
      Units_Sold   20
      Revenue_USD  996
      dtype: int64
```

```
[ ]:
```

4 Checking the shape of the data

```
[ ]: ### The data has 1000 rows &
```

```
[94]: df.shape
```

```
[94]: (1000, 10)
```

```
[ ]:
```

5 Check information of the data

5.0.1 Date needs to be changed from string to date data type

```
[7]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   Sale_ID         1000 non-null   str    
1   Date            1000 non-null   str    
2   Brand           1000 non-null   str    
3   Shoe_Type       1000 non-null   str    
4   Color           1000 non-null   str    
5   Country         1000 non-null   str    
6   Sales_Channel   1000 non-null   str    
7   Price_USD       1000 non-null   float64
8   Units_Sold      1000 non-null   int64   
9   Revenue_USD     1000 non-null   float64
dtypes: float64(2), int64(1), str(7)
memory usage: 78.3 KB
```

```
# View first 50 rows
```

```
[14]: df.head(50)
```

```
[14]:
```

	Sale_ID	Date	Brand	Shoe_Type	Color	Country \
0	S1	2025-11-24	Nike	Boots	Blue	UK
1	S2	2025-03-13	Skechers	Boots	Grey	USA
2	S3	2025-08-05	Nike	Running	White	UK
3	S4	2025-11-05	New Balance	Casual	Green	UAE
4	S5	2025-10-07	Adidas	Formal	Grey	France
5	S6	2025-03-23	Puma	Sports	White	UK
6	S7	2025-02-22	Reebok	Running	Blue	Saudi Arabia
7	S8	2025-11-06	Nike	Boots	Red	India
8	S9	2025-07-13	New Balance	Sports	Green	India
9	S10	2025-10-23	Skechers	Running	Black	UAE
10	S11	2025-05-29	Adidas	Running	Red	France
11	S12	2025-11-22	Adidas	Sports	Blue	UK
12	S13	2025-05-17	New Balance	Boots	White	India
13	S14	2025-05-06	Reebok	Formal	Blue	UAE
14	S15	2025-10-13	Skechers	Sports	Black	UK
15	S16	2025-06-11	Puma	Running	White	India
16	S17	2025-06-11	Skechers	Formal	Red	UAE
17	S18	2025-03-15	Adidas	Casual	Green	India
18	S19	2025-05-15	Reebok	Sneakers	Red	France

19	S20	2025-03-12	Reebok	Running	Black	Saudi Arabia
20	S21	2025-03-20	Skechers	Formal	Grey	USA
21	S22	2025-07-15	Reebok	Sneakers	Blue	India
22	S23	2025-12-15	Skechers	Sneakers	Blue	Saudi Arabia
23	S24	2025-06-24	Puma	Formal	White	Germany
24	S25	2025-05-15	Adidas	Sneakers	Black	Saudi Arabia
25	S26	2025-06-02	New Balance	Casual	White	France
26	S27	2025-10-04	Nike	Sneakers	Blue	Germany
27	S28	2025-02-27	Puma	Casual	Black	UK
28	S29	2025-02-10	Skechers	Formal	Black	Saudi Arabia
29	S30	2025-03-06	Skechers	Formal	Grey	UK
30	S31	2025-09-28	Reebok	Casual	Grey	Saudi Arabia
31	S32	2025-12-20	Skechers	Sports	Red	UAE
32	S33	2025-07-11	New Balance	Formal	Black	UK
33	S34	2025-02-02	Nike	Sneakers	Grey	UK
34	S35	2025-04-23	Nike	Boots	Green	USA
35	S36	2025-02-04	Puma	Running	Grey	UK
36	S37	2025-12-09	Adidas	Sneakers	White	UAE
37	S38	2025-10-23	Adidas	Formal	Red	UK
38	S39	2025-02-19	Puma	Formal	Red	Germany
39	S40	2025-01-28	Nike	Formal	Green	France
40	S41	2025-05-08	Adidas	Sneakers	Red	UK
41	S42	2025-04-04	Reebok	Casual	Black	Germany
42	S43	2025-02-20	Skechers	Sneakers	Black	USA
43	S44	2025-03-27	Reebok	Formal	White	Saudi Arabia
44	S45	2025-01-31	Reebok	Running	Red	France
45	S46	2025-05-27	Skechers	Boots	Grey	UAE
46	S47	2025-09-07	Adidas	Sports	White	USA
47	S48	2025-10-05	Skechers	Sports	Black	USA
48	S49	2025-09-02	New Balance	Casual	Black	India
49	S50	2025-04-06	New Balance	Running	Green	Saudi Arabia

	Sales_Channel	Price_USD	Units_Sold	Revenue_USD
0	Online	112.40	4	449.60
1	Mall	239.16	4	956.64
2	Mall	191.04	2	382.08
3	Mall	161.70	1	161.70
4	Online	64.32	14	900.48
5	Retail Store	64.32	14	900.48
6	Retail Store	42.78	3	128.34
7	Online	220.56	9	1985.04
8	Retail Store	162.25	3	486.75
9	Online	185.78	7	1300.46
10	Retail Store	34.53	3	103.59
11	Mall	243.38	12	2920.56
12	Mall	213.14	3	639.42
13	Mall	76.71	6	460.26

14	Online	70.00	8	560.00
15	Mall	70.35	13	914.55
16	Retail Store	96.93	7	678.51
17	Mall	145.45	9	1309.05
18	Online	125.03	19	2375.57
19	Online	94.07	17	1599.19
20	Retail Store	164.61	6	987.66
21	Online	60.69	20	1213.80
22	Mall	94.27	4	377.08
23	Online	110.60	4	442.40
24	Mall	130.34	17	2215.78
25	Online	202.74	17	3446.58
26	Online	73.93	17	1256.81
27	Mall	143.13	12	1717.56
28	Mall	160.33	3	480.99
29	Retail Store	40.22	5	201.10
30	Mall	163.66	20	3273.20
31	Mall	67.52	7	472.64
32	Online	44.31	15	664.65
33	Mall	238.75	11	2626.25
34	Online	242.44	1	242.44
35	Retail Store	207.85	2	415.70
36	Mall	97.02	16	1552.32
37	Online	51.49	16	823.84
38	Mall	180.53	14	2527.42
39	Online	126.83	4	507.32
40	Retail Store	56.85	7	397.95
41	Mall	138.94	9	1250.46
42	Online	37.57	2	75.14
43	Retail Store	230.05	14	3220.70
44	Retail Store	86.93	6	521.58
45	Mall	175.75	14	2460.50
46	Mall	98.58	5	492.90
47	Mall	144.41	2	288.82
48	Online	150.28	17	2554.76
49	Online	70.67	3	212.01

View last 50 rows

```
[19]: df.tail(50)
```

```
[19]:
```

	Sale_ID	Date	Brand	Shoe_Type	Color	Country \
950	S951	2025-10-05	Adidas	Sneakers	White	India
951	S952	2025-06-15	Puma	Sneakers	Grey	UK
952	S953	2025-09-29	Reebok	Sports	Blue	India
953	S954	2025-08-21	Skechers	Sneakers	Red	UAE
954	S955	2025-01-10	Puma	Sneakers	White	France

955	S956	2025-02-02	Reebok	Casual	Grey	UAE
956	S957	2025-10-10	New Balance	Sports	Blue	Germany
957	S958	2025-06-14	Nike	Sneakers	Green	USA
958	S959	2025-03-08	Nike	Casual	Black	Germany
959	S960	2025-02-20	New Balance	Boots	Red	UAE
960	S961	2025-09-27	Skechers	Sneakers	Blue	USA
961	S962	2025-04-24	Skechers	Running	Black	UK
962	S963	2025-09-09	Reebok	Boots	Black	France
963	S964	2025-08-01	Skechers	Running	Grey	France
964	S965	2025-01-21	Puma	Running	White	India
965	S966	2025-02-21	New Balance	Formal	White	UK
966	S967	2025-12-03	New Balance	Boots	White	India
967	S968	2025-10-13	Skechers	Running	Green	India
968	S969	2025-12-12	Skechers	Boots	Red	USA
969	S970	2025-04-03	Reebok	Sneakers	Blue	UAE
970	S971	2025-10-01	Reebok	Sports	Blue	India
971	S972	2025-07-10	Skechers	Formal	Green	France
972	S973	2025-05-04	New Balance	Casual	Red	Saudi Arabia
973	S974	2025-05-11	Reebok	Formal	Red	USA
974	S975	2025-01-18	New Balance	Running	Grey	UK
975	S976	2025-01-04	Adidas	Casual	Blue	USA
976	S977	2025-05-22	New Balance	Sneakers	Grey	India
977	S978	2025-02-18	Reebok	Sports	White	UK
978	S979	2025-07-07	Adidas	Casual	Blue	Saudi Arabia
979	S980	2025-02-17	Reebok	Sneakers	White	UAE
980	S981	2025-06-16	Adidas	Formal	Green	UAE
981	S982	2025-02-18	Skechers	Sports	Black	UAE
982	S983	2025-07-15	Nike	Sports	Blue	Saudi Arabia
983	S984	2025-08-02	Adidas	Boots	Green	Saudi Arabia
984	S985	2025-01-06	Nike	Sports	Grey	USA
985	S986	2025-10-04	Adidas	Sports	Grey	UAE
986	S987	2025-03-20	Skechers	Boots	Red	Germany
987	S988	2025-10-22	Nike	Casual	Red	France
988	S989	2025-03-17	Puma	Running	Red	India
989	S990	2025-01-26	Reebok	Sports	Black	Germany
990	S991	2025-08-11	Adidas	Formal	Grey	UK
991	S992	2025-07-22	Adidas	Sneakers	Grey	UAE
992	S993	2025-05-07	Nike	Sports	Red	France
993	S994	2025-10-22	Nike	Boots	Red	Germany
994	S995	2025-06-28	Puma	Sports	Grey	UAE
995	S996	2025-06-06	Puma	Sneakers	Red	India
996	S997	2025-08-14	Adidas	Casual	Green	France
997	S998	2025-02-13	Skechers	Running	White	India
998	S999	2025-05-28	Puma	Boots	Black	France
999	S1000	2025-08-30	Puma	Casual	Green	India

Sales_Channel Price_USD Units_Sold Revenue_USD

950	Retail Store	162.11	16	2593.76
951	Online	176.31	15	2644.65
952	Retail Store	68.58	5	342.90
953	Online	231.17	1	231.17
954	Mall	122.13	20	2442.60
955	Retail Store	114.29	18	2057.22
956	Mall	144.16	5	720.80
957	Online	40.33	4	161.32
958	Online	66.58	6	399.48
959	Retail Store	192.37	2	384.74
960	Retail Store	48.22	20	964.40
961	Mall	162.69	20	3253.80
962	Mall	83.98	3	251.94
963	Mall	115.65	12	1387.80
964	Online	93.51	3	280.53
965	Mall	108.25	8	866.00
966	Mall	188.19	8	1505.52
967	Retail Store	95.37	14	1335.18
968	Online	154.61	15	2319.15
969	Online	134.73	18	2425.14
970	Retail Store	176.01	20	3520.20
971	Mall	236.10	14	3305.40
972	Mall	191.17	19	3632.23
973	Mall	77.29	12	927.48
974	Retail Store	36.86	10	368.60
975	Mall	87.70	10	877.00
976	Online	160.92	18	2896.56
977	Retail Store	41.31	2	82.62
978	Retail Store	139.20	19	2644.80
979	Online	161.31	19	3064.89
980	Mall	103.53	16	1656.48
981	Mall	199.60	20	3992.00
982	Retail Store	53.45	3	160.35
983	Mall	46.53	4	186.12
984	Retail Store	190.20	13	2472.60
985	Online	139.01	19	2641.19
986	Mall	181.45	2	362.90
987	Retail Store	125.66	17	2136.22
988	Mall	84.21	5	421.05
989	Mall	210.20	18	3783.60
990	Retail Store	205.87	5	1029.35
991	Mall	182.83	19	3473.77
992	Mall	89.87	8	718.96
993	Retail Store	159.85	3	479.55
994	Online	109.41	7	765.87
995	Online	50.15	6	300.90
996	Online	231.81	2	463.62

997	Online	60.10	4	240.40
998	Retail Store	239.05	1	239.05
999	Online	128.12	1	128.12

6 Change date data type from string to date data type

```
[17]: df['Date'] = pd.to_datetime(df['Date'], errors = 'coerce')
```

6.0.1 Date has been changed to datetime data type

```
[22]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Sale_ID         1000 non-null   str
1   Date            1000 non-null   datetime64[us]
2   Brand           1000 non-null   str
3   Shoe_Type       1000 non-null   str
4   Color           1000 non-null   str
5   Country         1000 non-null   str
6   Sales_Channel   1000 non-null   str
7   Price_USD       1000 non-null   float64
8   Units_Sold      1000 non-null   int64
9   Revenue_USD     1000 non-null   float64
dtypes: datetime64[us](1), float64(2), int64(1), str(6)
memory usage: 78.3 KB
```

7 View Columns

```
[25]: df.columns
```

```
[25]: Index(['Sale_ID', 'Date', 'Brand', 'Shoe_Type', 'Color', 'Country',
          'Sales_Channel', 'Price_USD', 'Units_Sold', 'Revenue_USD'],
          dtype='str')
```

8 “Comparative Analysis of Revenue Across Sales Channels”

The analysis below explores how revenue varies across different sales channels Mall, Online, and Retail Store. By grouping the data based on the sales channel and calculating the average revenue, a clear pattern emerges. The Retail Store channel stands out as the top performer, generating the highest average revenue of approximately 1493 USD. This suggests that customers in physical retail environments tend to spend more, possibly due to in-store experiences, promotions, or product

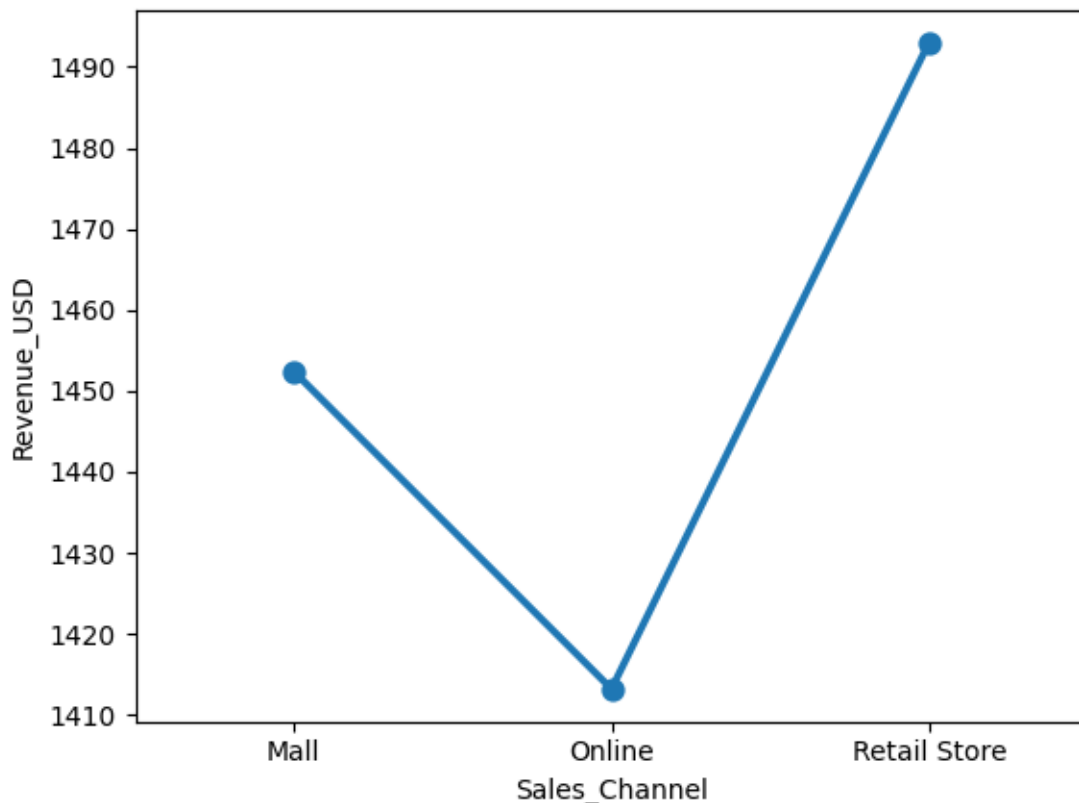
visibility. The Mall channel follows with an average revenue of about 1452 USD, indicating stable performance and consistent customer spending. Malls continue to play an important role in driving sales, although not as strongly as standalone retail stores. In contrast, the Online channel records the lowest average revenue at around 1413 USD. This may reflect lower transaction values online, possibly due to price comparisons, discounts, or less impulse buying compared to physical locations. The point plot visually confirms this trend, showing a noticeable dip for Online sales and a peak for Retail Store sales.

```
[42]: df2= df.groupby('Sales_Channel')['Revenue_USD'].mean()  
df2
```

```
[42]: Sales_Channel  
Mall          1452.389636  
Online        1413.209426  
Retail Store  1492.994484  
Name: Revenue_USD, dtype: float64
```

```
[43]: sns.pointplot(df2)
```

```
[43]: <Axes: xlabel='Sales_Channel', ylabel='Revenue_USD'>
```



```
[ ]:
```

9 “Daily Revenue Trends Analysis Over Time”

The analysis examines how total revenue changes over time across the year by aggregating daily sales. By grouping revenue by date and visualizing it using a line plot, a dynamic picture of business performance over time is revealed. At the beginning of the year, revenue shows moderate activity, with some days performing better than others. As time progresses, the pattern becomes more volatile, with sharp peaks and dips throughout the year. These fluctuations indicate that sales are not consistent on a daily basis and are influenced by varying factors such as demand, promotions, or seasonal effects. Notably, there are several significant spikes, where revenue reaches unusually high levels (close to 20,000 USD at some points). These peaks may correspond to special events, promotional campaigns, or high-demand periods such as holidays or sales seasons. On the other hand, there are also low-revenue days, where sales drop significantly, suggesting periods of weak demand or limited business activity. Overall, the trend does not follow a smooth upward or downward direction but instead shows high variability, indicating that the business experiences frequent changes in daily performance rather than steady growth.

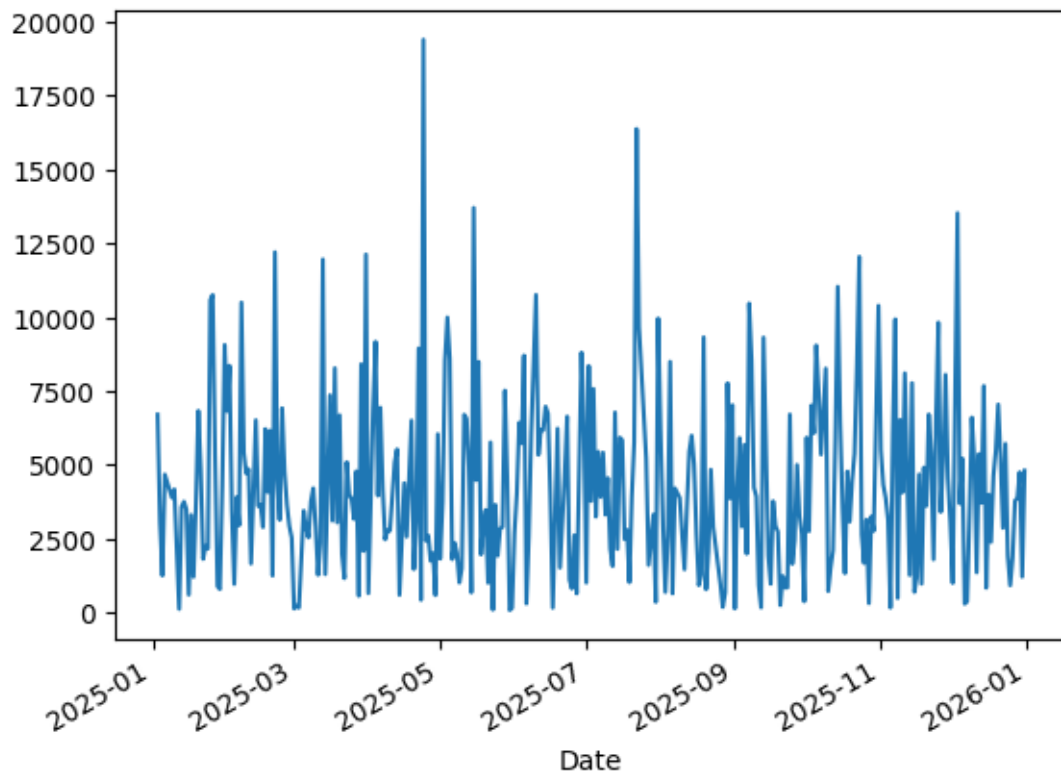
```
[61]: df4= df.loc[:, ['Date', 'Revenue_USD']]
```

```
[63]: df4= df4.groupby('Date')['Revenue_USD'].sum()  
df4
```

```
[63]: Date  
2025-01-03    6700.11  
2025-01-04    3039.76  
2025-01-05    1245.83  
2025-01-06    4670.12  
2025-01-09    3882.32  
...  
2025-12-27    3779.92  
2025-12-28    3870.04  
2025-12-29    4749.52  
2025-12-30    1208.18  
2025-12-31    4799.52  
Name: Revenue_USD, Length: 342, dtype: float64
```

```
[65]: df4.plot.line()
```

```
[65]: <Axes: xlabel='Date'>
```



[]:

Box Plot Summary (Revenue Distribution)

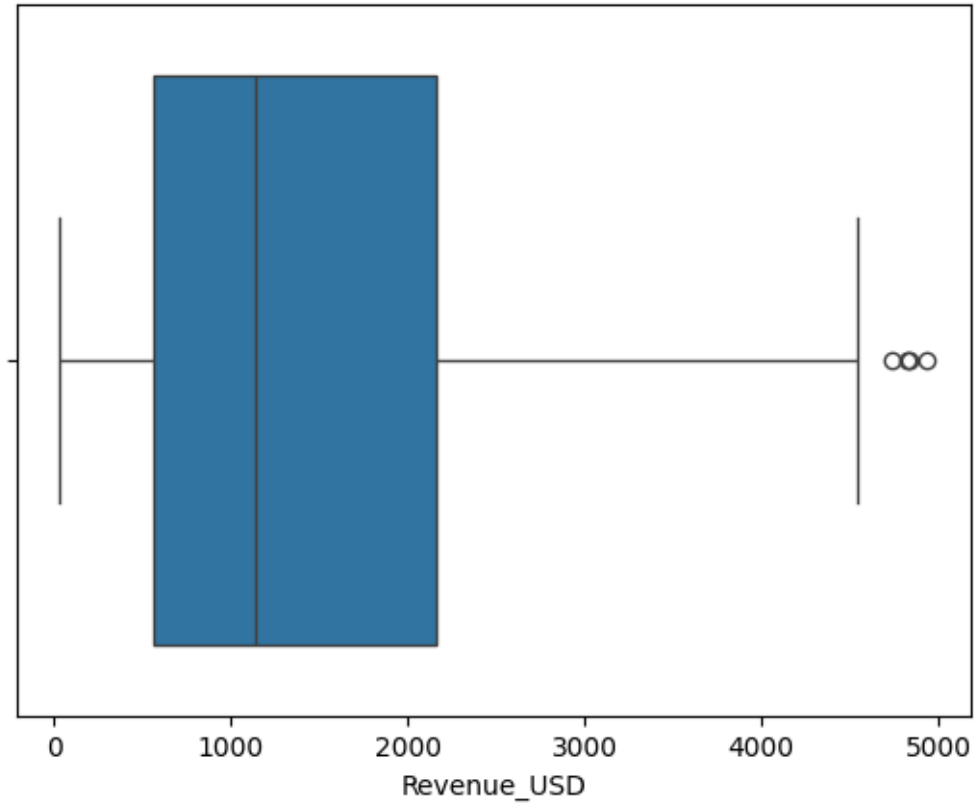
- 9.0.1 The box plot provides a clear view of how Revenue_USD is distributed across all observations in the dataset. The data shows a wide spread of revenue values, indicating significant variability in sales performance.
- 9.0.2 The median revenue appears to be around 1200 USD, meaning that half of the transactions fall below this value and half above it. The central box (interquartile range) stretches roughly from 700 USD to 2200 USD, showing where the middle 50% of revenue values lie.
- 9.0.3 The whiskers extend from near 0 USD up to around 4500 USD, indicating the overall range of most data points. However, there are a few outliers above 4500 USD, reaching close to 5000 USD, which represent unusually high revenue values compared to the rest of the data.
- 9.0.4 This distribution suggests that while most sales fall within a moderate range, there are occasional high-value transactions that significantly increase total revenue.
- 9.0.5 Key Insight:
- 9.0.6 The revenue data is right-skewed, meaning most values are lower, with a few high-value outliers driving peaks in performance.

```
[69]: df5= df.loc[:, ['Revenue_USD']]  
df5
```

```
[69]: Revenue_USD  
0      449.60  
1      956.64  
2      382.08  
3      161.70  
4      900.48  
..      ...  
995     300.90  
996     463.62  
997     240.40  
998     239.05  
999     128.12  
  
[1000 rows x 1 columns]
```

```
[80]: sns.boxplot(df5, x = 'Revenue_USD')
```

```
[80]: <Axes: xlabel='Revenue_USD'>
```



[]:

10 Box Plot Summary (Revenue Distribution)

The analysis examines how units sold vary across different brands and shoe types, providing insight into product demand and brand performance.

Overall, Sneakers and Running shoes emerge as the most popular categories across multiple brands, indicating strong customer preference for comfortable and casual footwear. For example, Adidas records its highest sales in Sneakers (438 units), while Skechers shows strong performance in Running shoes (398 units), highlighting dominance in athletic and lifestyle segments.

Looking at brand performance, Adidas and New Balance consistently achieve high sales across most categories, showing a well-balanced product demand. Adidas leads significantly in Sneakers, while New Balance performs strongly in Boots (386 units) and Formal shoes (351 units), suggesting versatility in both casual and formal markets. Nike shows relatively lower sales in some categories, particularly in Running (181 units) and Sports (170 units), which may indicate weaker performance or competition in those segments. However, Nike still maintains solid performance in Sneakers (354 units).

Puma demonstrates a balanced performance across all categories, with particularly strong results in Formal (354 units) and Sports shoes (307 units), suggesting a diverse customer base. Reebok

and Skechers also contribute strong sales, especially in Casual, Running, and Sneakers categories, indicating consistent demand in everyday and active footwear.

Key Insight: Customer demand is highest for casual and athletic footwear (Sneakers and Running shoes), and brands that perform well across multiple categories tend to achieve stronger overall sales.

```
[82]: df.groupby(['Brand', 'Shoe_Type'])['Units_Sold'].sum()
```

```
[82]: Brand      Shoe_Type  Units_Sold
      Adidas      Boots        329
           Casual        252
           Formal        291
           Running       305
           Sneakers      438
           Sports       238
      New Balance Boots        386
           Casual        286
           Formal        351
           Running       276
           Sneakers      324
           Sports       219
      Nike      Boots        264
           Casual        231
           Formal        271
           Running       181
           Sneakers      354
           Sports       170
      Puma      Boots        298
           Casual        269
           Formal        354
           Running       243
           Sneakers      256
           Sports       307
      Reebok     Boots        246
           Casual        370
           Formal        304
           Running       256
           Sneakers      283
           Sports       241
      Skechers   Boots        343
           Casual        307
           Formal        301
           Running       398
           Sneakers      324
           Sports       318
      Name: Units_Sold, dtype: int64
```

```
[ ]:
```

11 Filter Price, Units Sold, Revenue Columns

```
[ ]: df6= df.loc[:, ['Price_USD', 'Units_Sold', 'Revenue_USD']]  
df6
```

12 Pair Plot Summary (Price, Units Sold, and Revenue)

The pair plot provides a comprehensive view of the relationships between Price_USD, Units_Sold, and Revenue_USD, highlighting both distributions and correlations.

The diagonal histograms show that Price_USD is fairly spread out across a wide range, suggesting products are sold at different price levels. Units_Sold appears evenly distributed, indicating that sales volumes are relatively consistent across transactions. In contrast, Revenue_USD is right-skewed, meaning most values are lower, with fewer high-revenue observations.

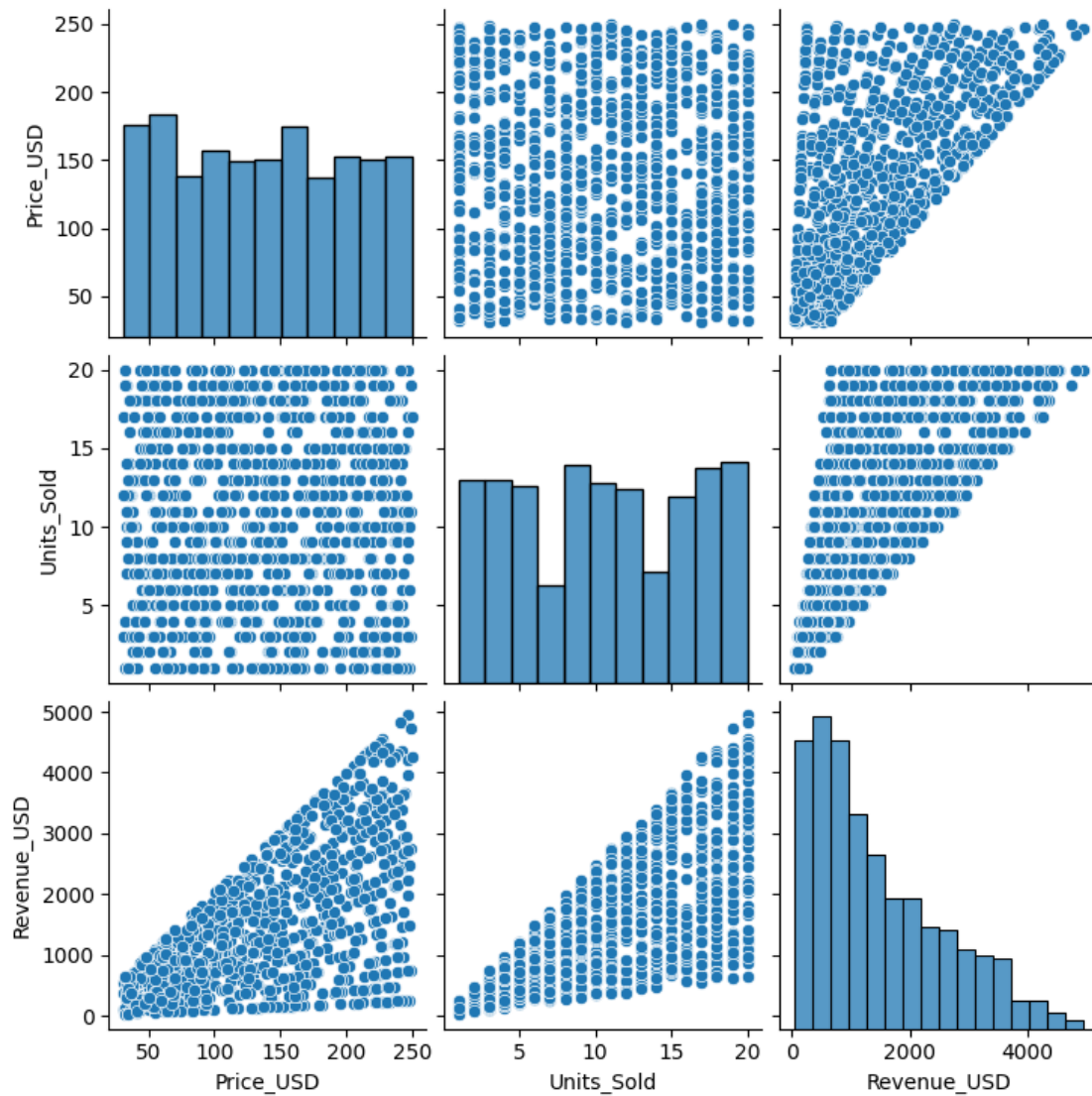
Looking at the scatter plots, there is a strong positive relationship between Units_Sold and Revenue_USD. As the number of units sold increases, revenue also increases significantly, which is expected since revenue is directly influenced by sales volume.

There is also a positive relationship between Price_USD and Revenue_USD, though it is more spread out. Higher-priced items can generate higher revenue, but the effect depends on how many units are sold. However, the relationship between Price_USD and Units_Sold appears weaker, suggesting that higher prices do not always lead to fewer sales, and customer demand is not strictly dependent on price alone.

Key Insight: Revenue is primarily driven by units sold, with price playing a secondary role. The combined effect of both variables determines overall sales performance, with high revenue achieved when both price and quantity are high.

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[89]: sns.pairplot(df6)
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[89]: <seaborn.axisgrid.PairGrid at 0x2461f546fd0>
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